

ROLE OF EARLY EXPOSURE TO INFLAMMATORY STRESS ON PLAQUE
DEPOSITION AND MICROGLIA DENSITY IN ALZHEIMER'S DISEASE USING THE
5XFAD MOUSE MODEL

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ABSTRACT

ROLE OF EARLY EXPOSURE TO INFLAMMATORY STRESS ON PLAQUE DEPOSITION AND MICROGLIA DENSITY IN ALZHEIMER'S DISEASE USING THE 5XFAD MOUSE MODEL

by Joshua Keene

Alzheimer's disease is a progressive neurodegenerative disease that causes memory loss, behavioral deficits and ultimately loss of normal bodily functions. Several variables contribute to Alzheimer's disease onset and progression, including protein dysfunction and aggregation. One type of protein dysfunction involves the peptide amyloid- β , which can accumulate to form neurotoxic oligomers and plaques. The process of neuroinflammation is strongly related to amyloidogenesis—likely in a bidirectional relationship. Microglia, the primary immune cells of the CNS and drivers of neuroinflammation, respond to amyloid- β through receptor-mediated phagocytosis and cytokine production. Prenatal exposure to pro-inflammatory cytokines is believed to shift microglia into a reactive state, often accompanied by microgliosis. These changes are likely to affect amyloidogenesis and amyloid- β removal during later development. To test this, male and female mice were bred from different maternal genotypes, derived from wild type or transgenic mothers. Transgenic mice harbor significant plaque deposition and reactive microglia expressing pro-inflammatory cytokines, therefore, mice born from transgenic mothers will undergo prenatal inflammatory stress. This study used the 5xFAD transgenic mouse model, which incorporates 3 human amyloid precursor protein mutations and 2 human presenilin 1 mutations, leading to early and aggressive amyloid- β and plaque formation. Mice exposed to prenatal inflammatory stress demonstrated significantly decreased hippocampal plaque deposition compared to mice not subjected to prenatal inflammatory stress. This phenomenon was observed in both sexes. However, prenatal inflammatory stress did not

significantly influence microglia density, regardless of sex. This suggests that efficiency of amyloid plaque removal, rather than quantity of microglia themselves, mediates amyloid pathology. Priming of microglia through exposure to inflammation during the asymptomatic phase of the disease, may account for this decrease in plaque burden.

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CHAPTER I

INTRODUCTION

Alzheimer's disease (AD) is the most prevalent form of dementia, afflicting an estimated 5.5 million individuals in the United States (Alzheimer's Association, 2017). Due to a disproportionately aging population and increased lifespan, the number of individuals in the United States with AD is predicted to rise to 13.8 million by 2050 (Herbert et al., 2013). Despite this widespread prevalence, the etiology of AD is currently unknown. Multiple variables likely contribute to the debilitating conditions experienced in this disease.

AD is a progressive neurodegenerative disease that ultimately leads to onset and progression of cognitive and behavioral deficits. Cognitive and behavioral deficits progress over time, including; memory loss, disorientation in time and place, poor judgment, changes in mood and behavior, personality changes, difficulties with language and difficulties performing daily activities of living. The hippocampus is affected during the early symptomatic stages of AD, and hippocampal-dependent episodic memory deficits due to synaptic dysfunction and neuronal loss are classic phenomena associated with AD progression (Gomez-Isla et al., 1996). The entorhinal cortex, a bi-directional hub between the hippocampus and cortical regions, displays amyloid- β ($A\beta$) plaque deposition during the prodromal stages of the disease. In the cortex, topographical distribution of plaques appears evenly distributed throughout the cerebrum in early to moderate stages of the disease (Arnold et al. 1991). Development of AD eventually leads to language deficits, characterized by difficulty understanding and producing speech. During later stages, motor function begins to decline, and basic autonomic functions decrease. The 5xFAD mouse model used in this experiment produces cortical and hippocampal $A\beta$ plaque formation and

hippocampal-dependent memory deficits (Jawhar et al., 2012). Because amyloidogenesis pathophysiology influences hippocampal and widespread cortical areas beginning early in the disease process, we focused on these brain regions in this study. The clinical symptoms of AD are well documented, however, the underlying variables that contribute to AD onset and progression are not fully understood.

A plethora of evidence over the past few decades suggests that AD pathophysiology is influenced by several variables. Synapse degeneration and cell death may be the most important correlates with cognitive and behavioral deficits associated with AD (Andrade-Moraes et al., 2013; Fagan et al., 2005). It is well established that microglia are necessary for synaptic plasticity and homeostasis— reactive microglia mediated pro-inflammatory cytokine production stimulates apoptotic cellular machinery (Liu et al., 2001). Inflammation induced reactivity of microglia also result in stripped neuronal synapses (Trapp et al., 2007). These findings implicate microglia as central to the phenomenon of neurodegeneration in AD, making them a potentially useful target and biomarker of the pathophysiology of this disease. Microglia are stimulated by A β over-production and aggregation, and influence A β in turn.

One of the physiological hallmarks of AD is the accumulation of monomeric and oligomeric A β peptides and plaques (Glenner & Wong, 1984). A β formation is one important physiological aspect of AD in humans. However, A β dysfunction alone may be neither necessary nor sufficient to cause AD. Nature provides us with a test of sufficiency—Individuals with a significant quantity of plaques can be cognitively normal, plaques are not enough to cause AD-like dementia (Knopman et al., 2003). It may be that patients exhibiting plaque deposition, but no clinical manifestations, are in the preclinical phase of the disease. Plaques are not necessary to

cause AD dementia, as human immunotherapy clinical trials using antibodies that target and stimulate removal of plaques demonstrate. Trials targeting removal of plaques fail to attenuate AD symptomatology based on clinical measurements of behavior and cognition (Wang et al., 2017). These interventions may occur too late in the disease progression, after downstream influences stemming from A β dysfunction already altered physiology beyond repair. The role A β dysfunction plays in AD onset/progression has yet to be demonstrably determined. A β dysfunction and accumulation in mouse models of the disease provide a useful tool for studying amyloidogenesis as one measure of diseaselike onset and progression, as well as the pro-inflammatory state A β and plaque deposition reliably elicit.

Genetic studies of humans with early-onset familial AD implicate amyloid precursor protein (APP) and A β 's role in AD onset and progression. Autosomal dominant gene mutations in APP and presenilin 1 and 2 (PSEN1/2) give rise to earlyonset familial AD—occurring before the age of 65. Pathogenic APP mutations increase the quantity of A β production by coding for an APP protein that is more susceptible to BACE1 cleavage. PSEN1 and PSEN2 are catalytic subunits of γ -secretase, mutations in these genes produce abnormally high levels of A β through increase in γ -secretase cleavage. Recently, an APP mutation was demonstrated to be protective against the sporadic form of AD. This sporadic form is characterized by late-onset of symptomatology, manifesting after the age of 65. Jonsson and colleagues (2012) found that this beneficial allele decreases risk of AD prevalence by up to 40%, based on a genomic-based epidemiological survey. In vitro analysis revealed that this mutation produces an APP protein that contains a sub-optimal BACE1 cleavage site, reducing the rate at which A β is produced. BACE1 deletion has also been shown to reduce neuronal loss and synaptic dysfunction in

5xFAD mice (Ohno et al., 2007). Toxic A β and amyloid plaque formation occurs when the normal physiological function of APP goes awry.

APP is highly conserved in mammalian species. This transmembrane glycoprotein may exert neurotrophic effects and is implicated in the modulation of synapses (Muller, Deller & Korte, 2017). α -secretases normally proteolyze APP at the N-terminal ectodomain (extracellular), producing secreted APP α (sAPP α). These sAPP α fragments function as neurotrophic factors and are involved in synaptic plasticity (Hasebe et al., 2013). Following α -secretase cleavage, γ -secretase cleaves a Cterminal (intracellular) section in the transmembrane region, producing a short peptide called p3 (Fig.1). P3 peptides are non-toxic and are eventually degraded. The formation of A β occurs when APP is cleaved by a family of secretases called beta amyloid cleaving enzymes (BACE1). Cleavage of the N-terminal ectodomain by BACE1 produces secreted APP β (sAPP β). Subsequent cleavage of the transmembrane Cterminal domain by γ -secretase produces an A β polypeptide of at least 38 to 43 amino acids in length (Kang et al., 1987) (Fig.1). These newly formed peptides originate as monomers, eventually forming soluble oligomers. Evidence suggests soluble oligomers exert neurotoxic effects, influencing both the extracellular and intracellular milieu (DiChiara et al., 2017). Some oligomers, such as A β 42, are predisposed to become extracellular insoluble fibrils. These fibrils can aggregate to form β -pleated sheets, and eventually accumulate into the primary constituent of amyloid plaques. Prolonged exposure to amyloid plaques induces neurite dystrophy and neurotoxicity in moderate to high concentration. This toxic effect has been demonstrated in vitro in rodent and human cell cultures (Pike et al., 1991; Deshpande et al., 2006). Plaque load also correlates with neurodegeneration in vivo in mice containing PSEN1 and APP mutations (Schmitz et al., 2004). The 5xFAD model used in this study predominantly produces

A β 42, smaller quantities of various other peptide lengths are produced—largely from the Swedish APP mutation, which also produces the more soluble Amyloid- β 40 (A β 40). The usefulness of this amyloidogenic model extends beyond the usefulness of plaque formation for quantification of AD-like onset/progression—the 5xFAD model reliably produces neuroinflammation, which plays a central role in human AD.

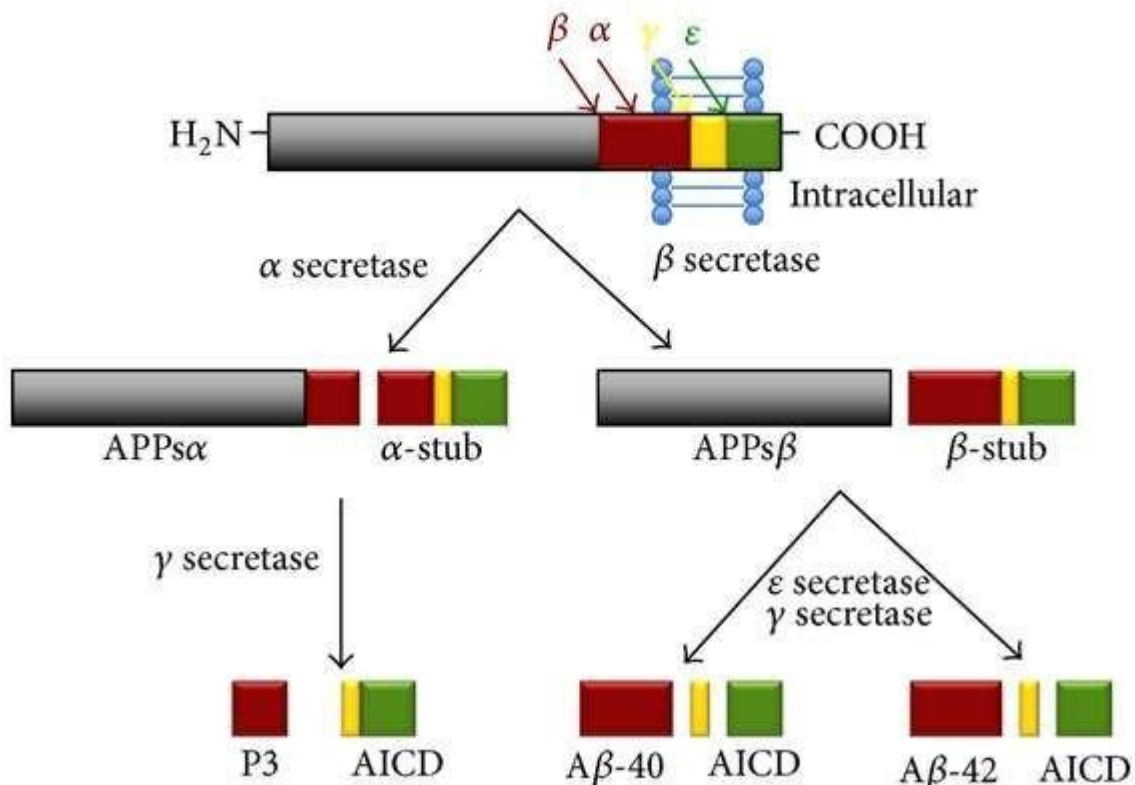


Figure 1. Schematic diagram depicting the processing of APP. α secretase (left) and β secretase (right), produce either sAPP α or sAPP β (respectively). This initial cleavage is followed by γ secretase cleavage of the C-terminal transmembrane region, producing either the non-toxic P3 (left) or neurotoxic A β peptide (right).

Note. Material adapted from “The Beta-Amyloid Protein of Alzheimer’s Disease: Communication Breakdown by Modifying the Neuronal Cytoskeleton” by Mokhtar et al., 2013, International Journal of Alzheimer’s Disease.

Epidemiological evidence in humans suggests that inflammation is significantly correlated with AD onset and progression. Epidemiological studies suggest that individuals who regularly use non-steroidal anti-inflammatory drugs (NSAID's) have a significantly lower incidence of AD compared to individuals who do not regularly use NSAID's (McGeer, Schulzer & McGeer, 1996; Stewart et al., 1997; Vlad et al., 2008). Human clinical trials tell a different story. Clinical trial results suggest that targeting inflammation does not significantly reduce AD progression in individuals with mild to moderate clinical symptoms. There have been several recent clinical trials targeting inflammation via NSAID administration—few of which have yielded significant results based on measures of AD clinical symptoms (Miguel-Alvarez et al., 2015). There are likely several reasons for this lack of findings, including the conflicting effects and often pleiotropic nature of these drugs. NSAID's are a diverse group of cyclooxygenase (COX) inhibitors that reduce prostaglandin production. Prostaglandin's can exert a wide range of downstream inflammatory influences, depending on the receptor they bind to. COX-2, for example, inhibits activation of prostaglandins that bind to the EP2 receptor. This receptor has a pro-inflammatory influence on microglia activity (Johansson et al., 2015). However, COX-2 also inhibits activation of the prostaglandin receptor EP4, a receptor with activity that induces anti-inflammatory effects in microglia (Shi et al., 2010). Drugs targeting COX-2 may have diametrically opposing actions simultaneously, depending on the prostaglandin receptor activated. This canvas approach towards dampening inflammation might contribute to the bewildering results from NSAID administration in clinical trials. A refined approach with selective targeting may be necessary, while being mindful of the impact that central nervous system (CNS) immune cells have at different stages of AD. Transgenic mouse studies targeting

inflammation through NSAID's generally yield beneficial results (McGeer and McGeer, 2007; Choi et al., 2013). Findings from these studies, as well as others, should be further clarified in the context of immune cell morphology and cytokine profile. CNS immunological function is a dynamic process in AD, dependent on the stage of the disease and the local cellular environment—mediated by microglia activity.

Microglia are the primary immune cells of the CNS. Under normal conditions, microglia act as sentinels in the brain, extending long filipodia that actively survey the extracellular fluid for abnormalities that indicate the presence of pathogens, toxins and cellular debris. In AD, the microglia phenotype can change from the typical surveying state to a ramified reactive state. During the reactive state, microglia are capable of migrating towards a perceived stressor—phagocytosing and degrading material. Like other monocyte derived immune cells, such as macrophages, microglia both respond to and produce a wide range of anti and pro-inflammatory cytokines that drive the inflammatory process in AD. Recent genetic studies shed light on the importance of microglia in the context of AD. Mutations in a gene labeled triggering receptor expressed on myeloid cells (TREM2) leads to significantly higher rates of AD in humans, and less containment of plaque area in mice (Yuon, et al., 2016). TREM2 is involved in microglia envelopment of A β and mediation of proper synaptic function in neurons. The range of microglia morphological phenotype and pro and antiinflammatory responses to A β is likely mediated by the surrounding milieu and cytokine activity (Mantovani et al., 2004; Colton et al., 2006; Sierra-Filardi et al., 2011).

Accompanying specific cytokine profiles is the phenomenon of microgliosis. Microgliosis is the proliferation of microglia in response to a stressor, in order to better deal with said stressor.

Gliososis is a reliable marker of a pro-inflammatory state, making microglia cell quantity a useful

marker of inflammation. In vivo positron emission tomography analysis in AD human patients shows that density of reactive microglia is significantly increased compared to healthy controls (Cagnin et al., 2001). Reactive microglia are present even in patients with mild AD-related dementia, providing evidence that microglia alterations occur early in the development of the disease. The temporal and spatial circumstances influencing microglia morphological alterations, gliosis, and diverse cytokine profile, may account for the seemingly conflicting and mixed results in human and animal studies of inflammation in AD. The capability of microglia-mediated A β removal is also influenced by the efficiency in which these cells phagocytose and degrade material.

Priming of microglia is one important aspect of microglia facilitated removal of A β and the chronic inflammatory state found in AD. Priming occurs when immune cells experience a stressor that produces an activated state. This activated state alters the gene expression signature of the cell, leading to a heightened response to subsequent stimuli (Holtman et al., 2015). In vitro analysis of microglia derived from cortical cell cultures of transgenic mice containing the PSEN1 and Swedish APP mutations demonstrate this priming effect—based on gene expression patterns and protein quantity of proteasome subunits (Orre et al., 2013). These proteasome subunits are necessary components of proteasome-mediated protein degradation necessary for removal of A β and other proteins during inflammation. Increased proteasome subunit transcription and protein synthesis occurs in cell cultures derived from human AD patients. Altered gene expression and protein synthesis based on these markers demonstrate that microglia are influenced by A β , not only in the short-term, but longterm as well. This suggests that early inflammatory stress may be a critical factor influencing AD onset and progression. In the current study, early inflammatory

stress imparted from transgenic mothers to their offspring prenatally may prime microglia, altering their subsequent response to A β formation.

Human epidemiological studies suggest that prenatal exposure to inflammatory stress is significantly correlated with several types of CNS disorders (Boksa, 2010). Rodent models mimicking prenatal inflammatory insult establish a clear transfer of inflammatory cytokines to fetal brains (Cai et al., 2000; Paintiliaet et al., 2004; Liverman et al., 2006). The pro-inflammatory cytokine profile of transgenic mice includes the production of: interleukin-1 α , interleukin-6, interleukin-12, interleukin-1 β , granulocytemacrophage colony stimulating factor, and TNF- α (Patel et al., 2005). Patel and colleagues showed that transgenic mice harboring PSEN1 and Swedish APP mutations display significantly higher levels of these pro-inflammatory cytokines. The 5xFAD transgenic mice used in the current study contain both of these human gene variants. This establishes a clear path of maternal inflammatory stress inheritance to progeny, and a potential priming effect. Factors that influence microglia morphology and function during the asymptomatic phase, should be examined as viable targets for AD therapeutics. It may be necessary to prevent or treat AD before physiology is beyond repair.

Sex differences may provide insight into the mechanisms involved in the relationship between neuroinflammation and AD pathophysiology. Evidence from human epidemiological studies suggests that females are more likely to be afflicted with AD, consisting of 60% of cases in the United States (Mufson et al., 2016). Further exacerbating this phenomenon, females also display a significantly faster progression of AD related cognitive decline compared to males, based on the well-established Braak stages (Braak & Braak, 1991). In humans, normal adult

female pre-menopause estrogen levels range from 15-350pg/ml, while normal adult males range from 1040pg/ml (Elmlinger et al., 2002). The wide-disparity of endogenous estrogen receptors between sexes, combined with higher circulating estrogen levels, may contribute to the increased rate of AD experienced in human post-menopausal females compared to males. One mechanism that may account for these between-sex differences is the role estrogen hormones play in neuroinflammation. 17 β -estradiol (E₂), the primary female sex hormone, exerts anti-inflammatory effects on microglia in mice (Vegeto et al., 2006). Vegeto et al. demonstrated that E₂ altered the reactive inflammatory state of microglia in response to lipopolysaccharide injection, marked by a decreased pro-inflammatory cytokine profile including reduction in TNF- α . This suggests a direct influence of estrogens on neuroinflammation—mediated by microglia. Clarifying sex-differences in microglia density and amyloid-pathology using stereological quantification will open the door for more mechanistic work in the future.

We hypothesize that exposure to prenatal inflammatory stress during the asymptomatic phase of AD will increase the inflammatory response later in development, lessening amyloid plaque burden. We further hypothesize that prenatal exposure to inflammation will demonstrate a sex-difference in microgliosis and amyloid plaque burden.

CHAPTER II

METHODS

All of the experiments carried out in this article were performed according to the National Institute of Health's Guidelines for the Care and Use of Laboratory Animals and approved by the IACUC committee at Central Michigan University in Mt. Pleasant, Michigan (protocol 15-01).

MICE

Male and female 5xFAD transgenic mice (Tg6799 line) were obtained from Jackson Laboratories (Maine, USA) harboring 3 human APP mutations; Swedish (K670N/M671L), Florida (I716V), and London (V717I) and 2 human PSEN1 mutations (M146L and L286V). Expression of these human transgenes is controlled by the Thy-1 promoter—which is reliably expressed post-natal day 7. The Florida, London and 2 PSEN1 mutations cause A β (1-42) production and accumulation while the Swedish mutation increases A β production non-specifically. These transgenes cause early and severe amyloid pathology beginning at 1.5 months of age. To obtain the lowest necessary number of animals for this study, we crossed 5xFAD females (up to 6 months of age) with B6/SJL wild type (WT) males. The Mendelian nature of the breeding paradigm provided roughly 50% male and 50% female litters, as well as 50% 5xFAD vs. WT animals. A total of 30 animals were used in this study. All mice were weaned on post-natal day 21 and housed in our on-site vivarium (Central Michigan University, HPB building) according to litter and sex. All animals had continual and full access to food and water, with a 12H inverted light/dark cycle.

Between 14 and 21 days after birth, each mouse was tail snipped for the acquisition of tissue for DNA analysis. Genotyping was performed using polymerase chain reaction. APP forward and reverse primers were used to identify 5xFAD animals because WT mice do not harbor APP genes. P35 was used for control primers—all mice contain the P35 gene involved in cell cycle.

TISSUE PREPARATION

Animals were deeply anesthetized with isoflurane via inhalation in a closed chamber. Mice were then transcardially perfused with 50ml of cold .9% sodium-chloride saline solution (pH 7.4), mixed with 1 IU/ml of Heparin sodium as an anticoagulant. 50ml of 4% para-formaldehyde (PFA) diluted in double-distilled H₂O (pH of 7.4) was then transcardially perfused through the circulatory system of the mouse to cross-link proteins—fixing the tissue. Both saline and 4% PFA were perfused using a Harvard Apparatus for a constant perfusion of fluids at a rate of 7 ml/minute. The brain was then carefully extracted from the skull and completely submerged in 4% PFA (pH 7.4) at 4°C for 24 hours for post-fixation. After 24 hours of post-fixation, the brain was removed from the PFA, rinsed twice with phosphate-buffered saline (PBS) (pH 7.4) and then stored in PBS at 4°C.

Brain tissue was cut into 30 µm thick coronal sections using a vibratome (Leica VT1000S). The brainstem and cerebellum were removed, and all viable forebrain tissue sections were kept, excluding the olfactory bulbs. Unfrozen tissue was sectioned in cold PBS (pH 7.4). Sectioned tissue was placed into 24 well plates, with 1.5 ml of cryoprotect containing ethylene glycol, glycerol, and .5M phosphate buffer (pH 7.4). Cryopreserved tissue was then stored at -20°C.

IMMUNOHISTOCHEMISTRY

Immunofluorescent staining was performed on 1/6th of tissue sections containing hippocampal and surrounding cortical regions starting at Bregma position -0.94 mm and ending at Bregma position -3.16 mm (anterior-posterior). Tissue was rinsed 3 times with PBS (pH 7.4) at room temperature (RT). Tissue was then incubated in a blocking buffer at RT for 1 hour and 30 minutes. The blocking buffer contained PBS (pH 7.4) with .1% triton x 100, 5% normal goat

serum, and 1% bovine serum albumin. Ionized calciumbinding adapter molecule 1 antibody (Iba1) was used as a marker of microglia activation. Tissue was incubated in anti-rabbit Iba1 polyclonal primary antibody (Wako, Richmond VA, USA). The antibody 6E10 recognizes the N-terminal region of A β peptides of both 40 and 42 amino acids in length. The A β 42 peptide shows a high affinity for accumulation, which unperturbed, ultimately results in plaque formation. Concurrently with Iba1, tissue was incubated with anti-mouse 6E10 polyclonal primary antibody (Biolegend, San Diego, CA, USA). Both primary antibodies were diluted to concentrations of 1:1000, and placed on a stir plate for 12 hours at 4°C. Tissue was then rinsed 3 times with PBS (pH 7.4) for 10 minutes at RT. Tissue was then incubated in secondary antibody, Alexa Fluor 488 goat anti-rabbit (for anti-Iba1) and Alexa Fluor 594 goat anti-mouse (for anti-6E10) (Invitrogen, Carlsbad, CA, USA) at a concentration of 1:500 for 2 hours and 30 minutes in blocking buffer at RT. Tissue was then rinsed 3 times with PBS (pH 7.4) and then counter-stained with Hoechst 33342 (ThermoFisher Scientific, Waltham, MA, USA) at a concentration of 1:2,000 for 5 minutes at RT. Hoechst dye stain shows specificity for adenine and thymine nucleotides of DNA, making it a useful marker for nuclei of all cells, including neurons and glial cells. Tissue was then rinsed 3 times with PBS (pH 7.4) at RT for 5 minutes and mounted on positively charged glass slides with Fluoromount-G mounting media (Southern Biotech, Birmingham, AL, USA). Stained and mounted tissue was stored at 4°C in the dark.

IMAGING AND STEREOLOGICAL ANALYSIS

Stained and mounted tissue sections were imaged using a Zeiss Axioimager M1 microscope. Hippocampal and surrounding cortical images were taken with a 20x objective for the purpose of stereology. 5 Z-plane steps (4 μ m apart) covering 20 μ m of tissue (all available

focal planes) were taken for all mounted tissue sections. All TG animals were imaged using 3 channels; Iba1 (488 nm), 6E10 (568 nm), and Hoechst (358 nm). TG animal images were processed for plaque area analysis using extended depth of focus, collapsing the Z-plane down into 1 focal plane—producing a 2dimensional image. Original full Z-plane images were also kept for quantification of microglia density. Because WT animals lack APP/Amyloid- β , only Iba1 (488 nm) and Hoechst (358 nm) channels were used for quantification of microglia and cell nuclei, respectively. The full Z-plane was kept for these images for microglia density analysis.

Unbiased stereology was performed for plaque area analysis using the Stereo Investigator system (MBF Bioscience, Williston, VT, USA). Because plaques are composed of accumulated peptides, they do not contain a unique identification point, unlike cells. Therefore, we measured plaque area percentage based on estimated plaque area divided by total area of the region of interest—either hippocampus or cortex. Plaque area was analyzed using the Area Fraction Fractionator probe for all TG mice. The parameters for this probe were set at 400 μm by 400 μm counting frame size, and 750 μm by 750 μm sampling grid size. Grid spacing was set at 130 μm , and probe height was set at 30 μm . All plaque types were included in this analysis, including those ranging from strict dense core plaques to diffuse type plaques. Only plaques directly touching crosshairs within the counting frame box were included.

Using the Stereo Investigator system, the Optical Fractionator Workflow probe was used to quantify microglia density. Microglia density was based on total estimated microglia population divided by total volume. Microglia density was converted into estimated population of microglia per mm^3 . Density analysis was performed for hippocampal regions of interest and cortical regions. The parameters for microglia quantification were set at 65 μm by 65 μm

counting frame box size, and 550 μm by 550 μm grid size. Section cut thickness was set at 30 μm .

STATISTICAL ANALYSIS

Independent samples unpaired T-tests were used for comparison of means related to sex differences based on the dependent variable of plaque area. To determine if maternal genotype played a role in plaque progression and microglia density, two-way ANOVA was performed for statistical comparisons of the independent variables sex and maternal genotype. Tukey's multiple comparisons post hoc test was used to determine if significant differences existed between groups. Alpha levels were set at .05.

CHAPTER III

RESULTS

HIPPOCAMPAL PLAQUE AREA

The effect of maternal genotype on hippocampal plaque area was significant within-sexes, in both male and female mice ($F=25.76$, $P=.0002$) (Fig.2D). TG males born from WT mothers showed significantly more plaque area in the hippocampus compared to TG males born from a TG mother (Fig.3), with a mean difference of 46.2% ($P=.026$). Female mice displayed this same phenomenon, on average, TG females born from WT mothers showed 58.5% greater hippocampal plaque area compared to TG females born from TG mothers ($P=.015$) (Fig.4). Based on a multiple comparisons twoway ANOVA, a significant sex-difference in plaque area percentage between male and female mice was demonstrated, with females exhibiting greater plaque area than males ($F=8.4$, $P=.011$).

HIPPOCAMPAL MICROGLIA DENSITY

Overall, hippocampal microglia density was significantly different between TG and WT animals within-sexes ($F=45.6$, $P<.0001$) (Fig.2A). Male TG mice showed 33.2% more microglia density than WT males ($P=.0022$) (Fig.3). Female TG mice showed 36.6% greater microglia density than WT females ($P=.0002$) (Fig.4). Female mice seem to present with greater microglia density in the hippocampus compared to males ($F=3.71$, $P=.071$) however, this disparity did not meet the criteria for statistical significance.

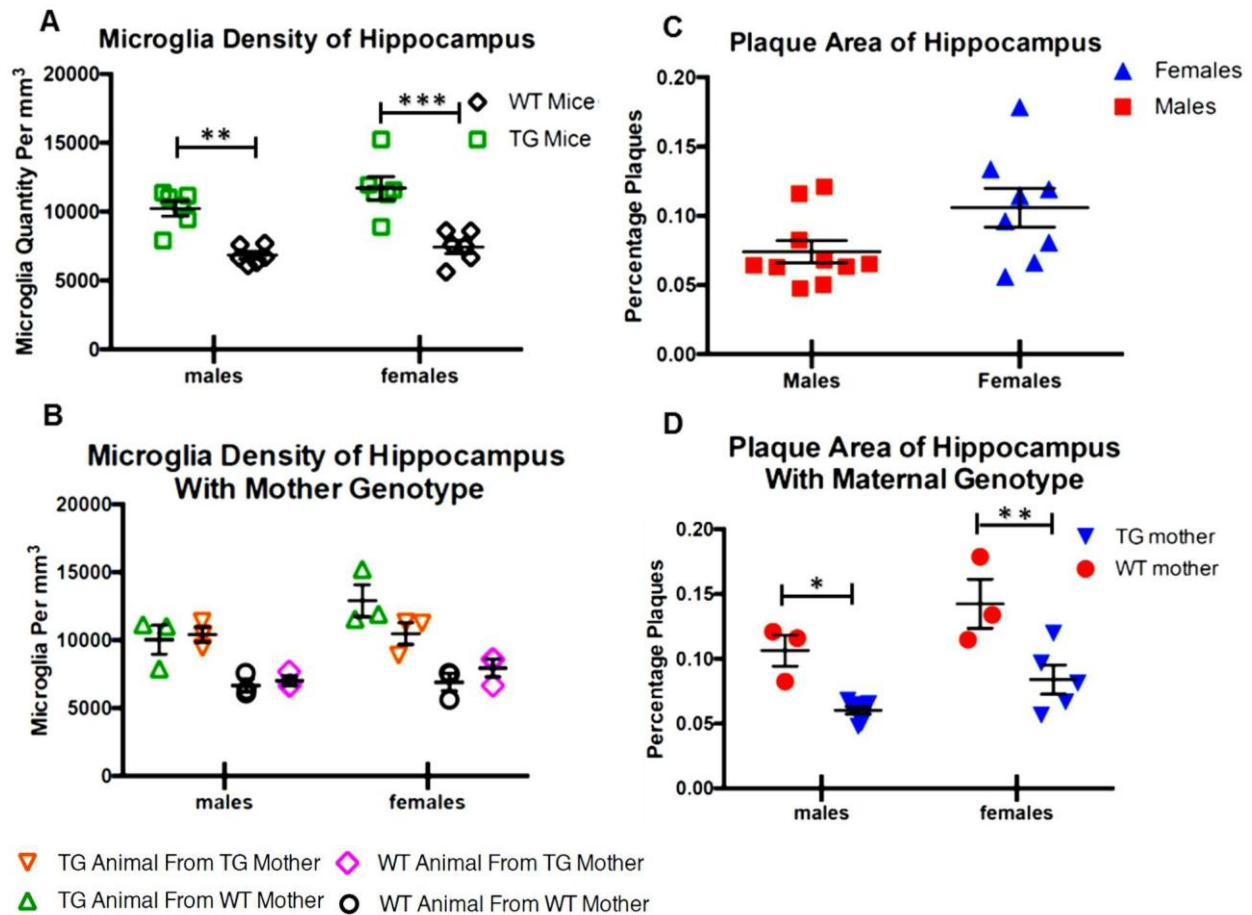


Figure 2. Stereological results of hippocampal plaque area and microglia density between sexes, genotype of the animal, and maternal genotypes.

(A) Microglia density of TG and WT animals between sexes. (B) Microglia density of TG and WT animals between sexes, including maternal genotype. (C) Plaque area percentage between

sexes, all of these animals are TG. (D) Plaque area percentage between sexes, including maternal genotype. All of these animals are TG.

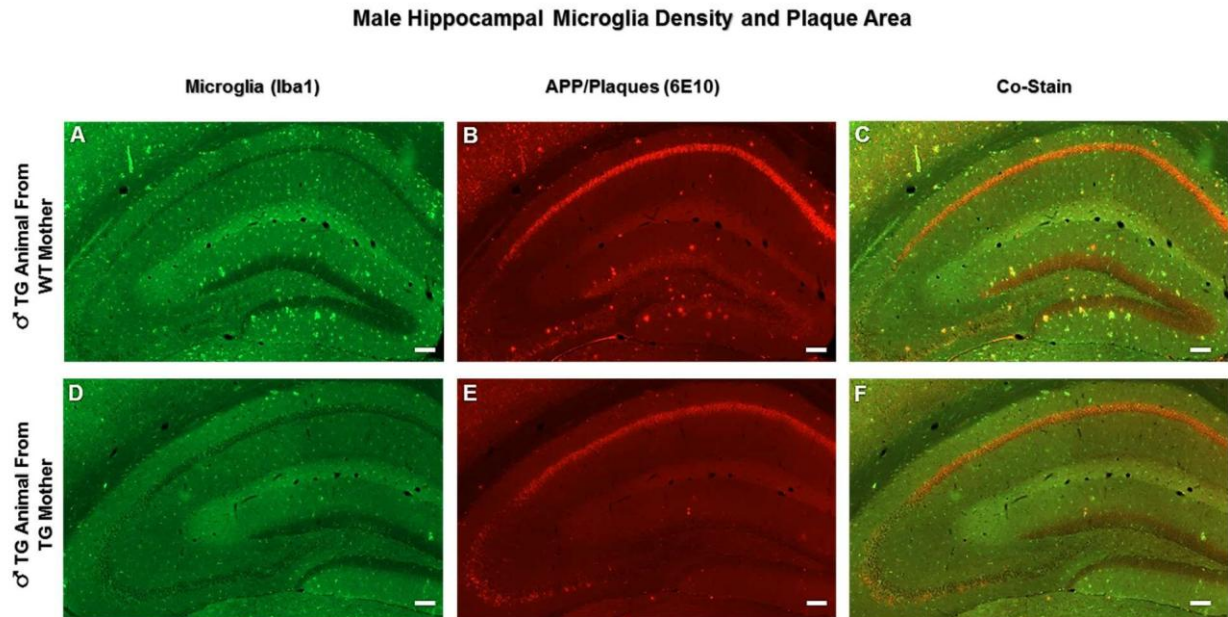


Figure 3. Male hippocampal microglia density and plaque area. Male TG animals born from WT mothers (top row) and male TG animals born from TG mothers (bottom row).

Key: Scale bar represents 200µm.

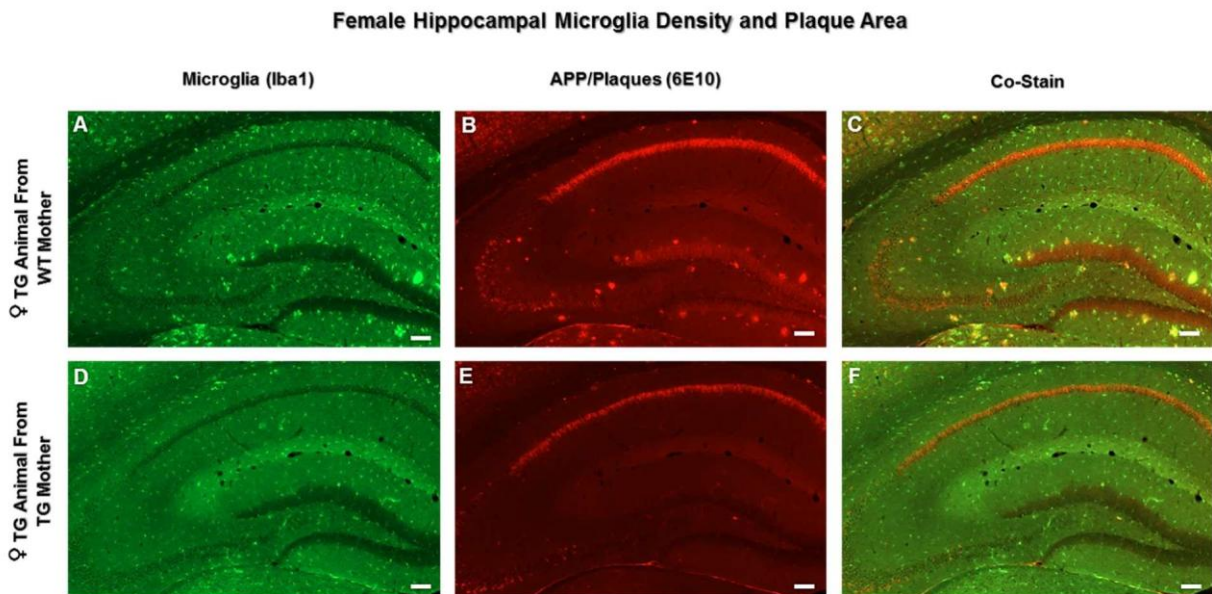


Figure 4. Female hippocampal microglia density and plaque area.
Female TG animals born from WT mothers (top row) and female TG animals born from TG mothers (bottom row).

Key: Scale bar represents 200 μ m.

CORTICAL PLAQUE AREA

There was a significant difference in cortical plaque area based on sex (Fig.3C).

Male mice displayed 24.8% less plaque area ($M=.0621$, $SEM=.0065$) than females

($M=.0905$, $SEM=.00825$) $t(16)=2.72$, $P=.0149$.

CORTICAL MICROGLIA DENSITY

Overall, there was a significant difference in cortical microglia density between animal genotypes ($F=25.92$, $P<.0001$) (Fig.3A). Female TG mice contained 31.74% greater microglia density than WT females ($P=.0009$) (Fig.7). Male TG mice showed 20.7% greater microglia density than their WT counterparts (Fig.6), however, this difference did not meet the criteria for statistical significance ($P=.074$).

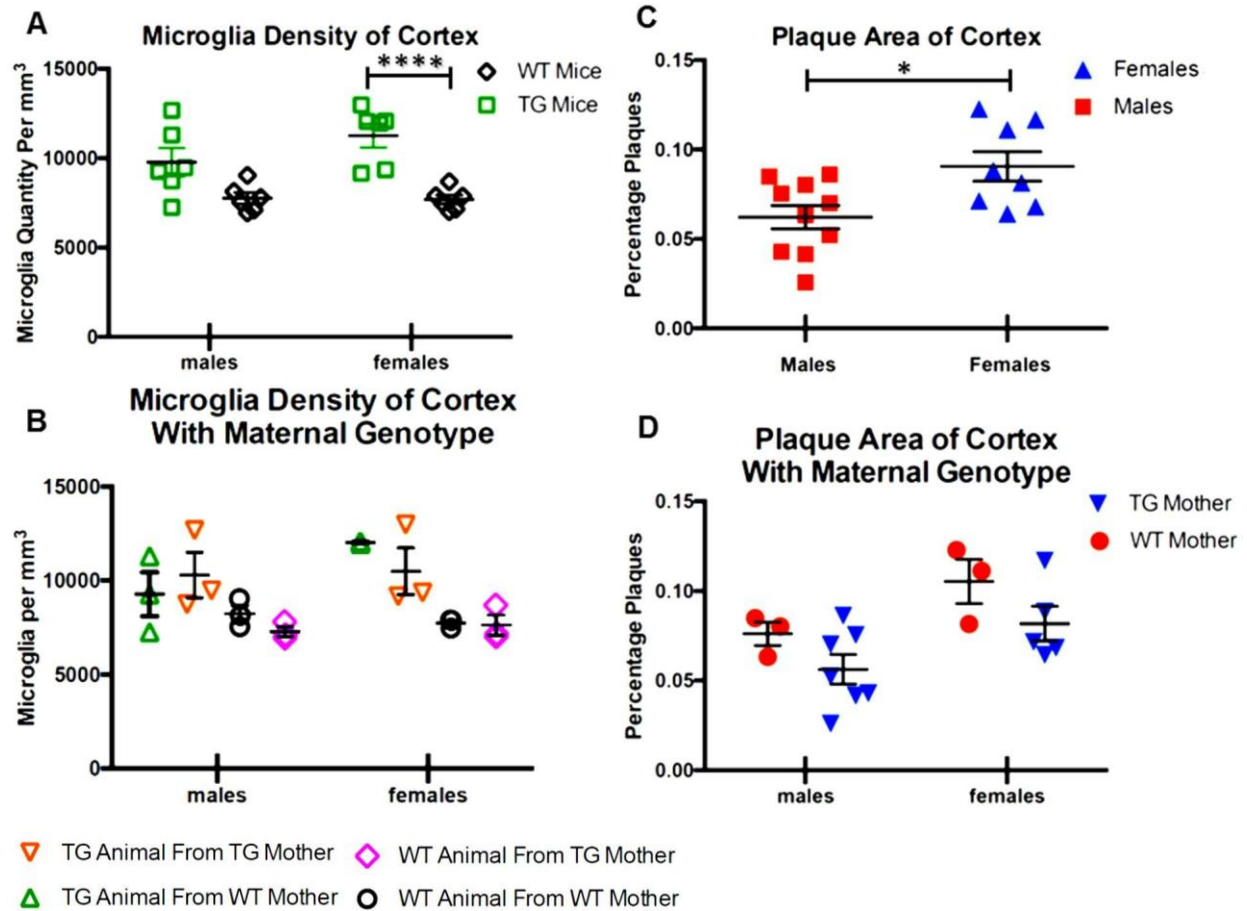


Figure 5. Stereological results of cortical plaque area and microglia density between sexes, genotype of the animal, and maternal genotypes.

(A) Microglia density of TG and WT animals between sexes. (B) Microglia density of TG and WT animals between sexes, including maternal genotype. (C) Plaque area percentage between sexes, all animals are TG (D) Plaque area percentage between sexes, including maternal genotype. All of these animals are TG.

Male Cortical Microglia Density and Plaque Area

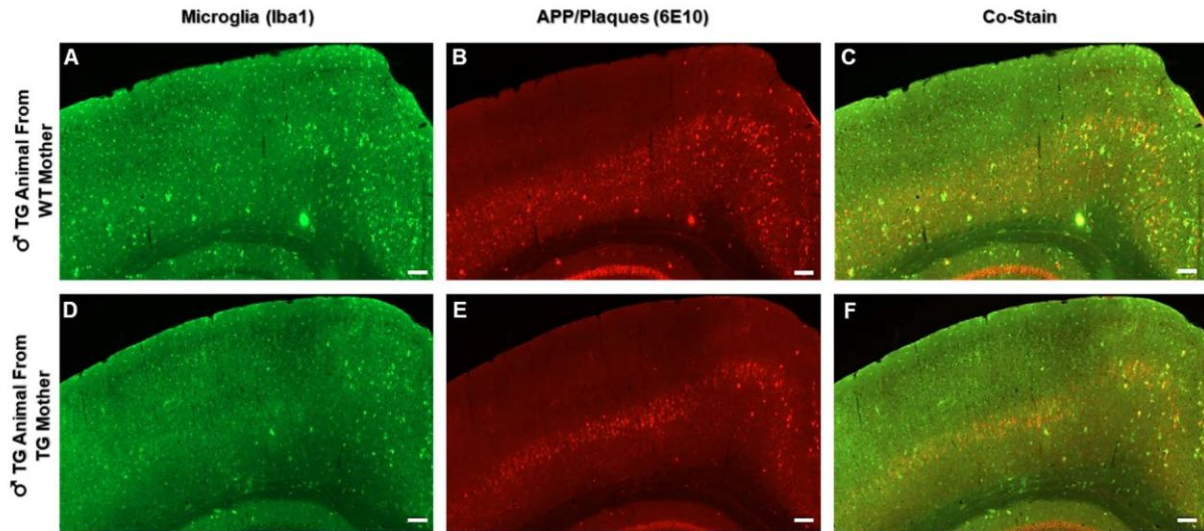


Figure 6. Male cortical microglia density and plaque area. Male TG animals born from WT mothers (top row) and male TG animals born from TG mothers (bottom row).

Key: Scale bar represents 200 μ m.

Female Cortical Microglia Density and Plaque Area

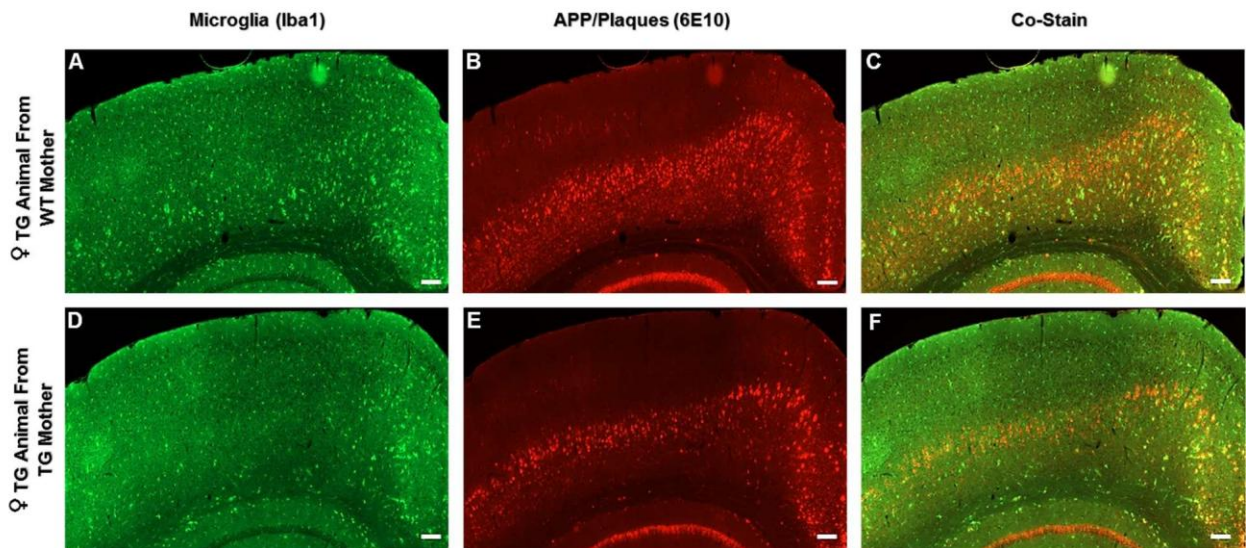


Figure 7. Female cortical microglia density and plaque area. Female TG animals born from WT mothers (top row) and female TG animals born from TG mothers (bottom row).

Key: Scale bar represents 200 μ m.

CHAPTER IV

DISCUSSION

Male and female mice exposed to prenatal inflammatory stress showed less plaque load compared to mice not exposed to prenatal immune stress, in both the cortex and hippocampus. Males exposed to inflammatory stress showed 46% less plaque area compared to animals not exposed to prenatal immune stress, while females showed 59% less plaque area in animals exposed to prenatal immune stress. These results suggest that reduced amyloidogenesis and/or removal of A β is occurring in animals that are exposed to early inflammatory stressors. This suggests a priming influence on microglia—prior inflammation increased the effectiveness of plaque removal and/or decreased plaque deposition. However, this was not correlated with an increase in microglia density, as no significant difference between exposure to prenatal stress and microglia density was found. These results are contrary to findings that microglia exposure to inflammatory stress significantly increases microgliosis.

A sex-difference was observed for plaque load in cortical regions, with females displaying greater plaque area. This phenomenon was also observed in the hippocampus but was not deemed statistically significant ($p=.055$). This between-sex phenomenon has been observed in previous TG mouse studies (Wang et al., 2003; Callahan et al., 2001). No sex-differences were observed between animals exposed to prenatal inflammation, suggesting that the influence of prenatal inflammatory stress did not significantly differ in females compared to males. The mechanism driving these overall sex differences in amyloidogenesis needs to be elucidated, most hypothesis link estrogen depletion after menopause as the driving force for between-sex differences in amyloid burden. Estrogen acts directly on microglia and is anti-inflammatory in the CNS (Vegeto et al., 2006). Mice in this study were examined at 4 months of age, which does

not correspond to typical cessation of menopause in female mice beginning at 8 months of age (Gosden et al., 1983). Callahan et al. (2001) demonstrated an increase in plaque area deposition in postmenopausal female mice at 15 and 19 months old, compared to age-matched males in animals harboring 3 APP mutations. Wang et al. (2003) demonstrated a similar finding, with no significant sex difference in mice at 4 months of age, but a significant sex difference at 12 and 17 months of age. However, Wang and colleagues study included only 1 APP mutation (Swedish) and 1 PSEN1 mutation—leading to the formation of plaques starting at 9 months of age. In contrast, the 5xFAD model displays marked plaque formation in the 4-month-old mice used in this study. This brings into question the overall usefulness of comparing results based on different mouse models of AD.

TG mice exhibited greater microglia density in both the hippocampus and cortex compared to WT mice. The APP and PSEN1 human genes responsible for the production of A β in the 5xFAD model are under the control of the Thy-1 promoter, which begins expression around post-natal day 7. Plaque accumulation, predominantly consisting of A β 42, begins to display around 2 months of age. Microglia in the 4-month-old TG mice used in this study would be exposed directly to A β stress—this is one suggested mechanism for the phenomenon of microgliosis observed in these TG mice. Microglia contain several cell-surface receptors capable of binding to A β oligomers and fibrils. One receptor capable of binding A β fibrils is the $\alpha_6\beta_1$ -integrin receptor, which associates with CD36 and CD47 to form an activated complex in the presence of A β fibrils (Bamberger et al., 2003). This $\alpha_6\beta_1$ -integrin receptor complex promotes adhesion of microglia to A β , pushing microglia towards a reactive pro-inflammatory phenotype. Upon A β insult, microglia also upregulate expression of the scavenger receptor A (SRA) and a receptor for advanced-glycosylation end-products (RAGE) during amyloidogenesis in

APP/PSEN1 TG mice (Suzanne et al., 2008). However, expression of these SRA and RAGE receptors in response to A β decreases with age upon chronic exposure to A β , suggesting a differential effectiveness of microglia at different stages of AD development. Microglia contain several other receptors capable of binding to A β oligomers and fibrils, including: toll-like receptor-2, -4, -6, -9, and cell surface receptor CD14 (Paresce, Ghosh & Maxfield, 1996; Bamberger et al., 2003; Liu et al., 2005; Stewart et al., 2010). Activation of pro-inflammatory type of microglia receptors is often followed by microgliosis—reactive microglia undergo cell division (Li and Zhang, 2016; Yates et al., 2000). The phenomenon of increased microgliosis in TG animals compared to their WT counterparts is well established. It is less clear what influence prenatal inflammatory stress has on progeny microgliosis.

Microgliosis was not observed based on exposure to prenatal inflammatory stress. This appears to conflict with the decreased plaque load in these same mice. These results also conflict with other TG mouse studies that show pro-inflammatory cytokine exposure induces microgliosis and removal of plaques (Chakrabarty et al., 2010). Ghosh and colleagues (2013) found a marked reduction in amyloid plaques and a 6x increase in microglia cell quantity after exposure to IL-1 β . IL-1 β is a proinflammatory cytokine found in animals expressing APP/PSEN1 mutations, such as the animals used in this study. Microglia density in female mice was greater in both TG and WT animals compared to males, although these statistics were not deemed significant. However, the increased microglia density in conjunction with increased plaque load in females suggests microglia cell quantity may not be the determining factor involved in plaque removal.

Overall, exposure to prenatal inflammatory stress during the asymptomatic phase of amyloidogenesis decreased plaque load in the hippocampus. This phenomenon was not sex-dependent, as both males and females exhibited a significant difference in plaque load based on maternal genotype. Conversely, prenatal exposure to inflammatory stress during the asymptomatic phase of amyloidogenesis did not significantly affect microglia density. These findings suggest that priming of microglia may occur independently of microgliosis. Microglia efficiency, rather than quantity, may significantly influence amyloid pathology. Cell-surface receptor alterations, in both quantity and function, may better represent the phagocytic and proteolytic capabilities of primed microglia. The predominant amoeboid morphological changes that microglia undergo during reactivity to pro-inflammatory stress may also be a more significant determining factor than microgliosis in terms of plaque clearance capabilities. Future studies should clarify the relationship between microglia cell density and phagocytic capabilities. This will require induction of inflammation before the onset and progression of the disease. Studies should also evaluate the mechanisms underlying sex differences found in plaque load percentage. A better understanding of the mechanisms driving these phenomena may provide useful targets for AD prevention and treatment.

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